COMPUTER NETWORKS

Unit IV, Unit V & Unit VI • Beginner-Friendly MCQ Exam Notes

How to study: Read the definition, understand the simple example, and memorize the MCQ points. ■■■■■ means a high-priority topic.

Quick idea of the three units

Network Layer: moves packets between networks. **Transport Layer:** delivers data to the correct application. **Application Layer:** provides services such as web, email, file transfer, and DNS.

UNIT IV — NETWORK LAYER PROTOCOLS

1. Network Layer Introduction ■■■■■

Definition: The Network Layer sends packets from one network to another. It handles logical addressing, routing, and forwarding.

Example: Like a GPS helping a letter travel from Surat to Delhi.

MCQ points:

• OSI Layer 3.

• Data unit: Packet.

• Routers operate at the Network Layer.

• IP is the main protocol.

2. IP – Internet Protocol ■■■■■

Definition: IP helps deliver packets from a source device to a destination across networks.

Example: A computer sends a packet to a website using the website's IP address.

MCQ points:

• IP is connectionless and best-effort.

• IP does not guarantee delivery or packet order.

• IPv4 address = 32 bits (4 bytes).

• IPv6 address = 128 bits (16 bytes).

• TCP can provide reliability above IP.

3. ICMPv4 ■■■■■

Definition: ICMPv4 sends error messages and diagnostic information for IPv4.

Example: The ping command uses ICMP Echo Request and Echo Reply to check reachability.

MCQ points:

• ICMP = Internet Control Message Protocol.

• Used for error reporting and diagnostics.

• Destination Unreachable and Time Exceeded are important messages.

• ICMP is carried inside IP packets.

4. Unicast Routing – Introduction ■■■■■

Definition: Unicast means sending data from one sender to one receiver. Routing chooses a path for packets.

Example: A laptop sends a request to one particular website.

MCQ points:

• Unicast = One-to-One.

• Multicast = One-to-selected-many.

5. Routing Algorithms ■■■■■

Definition: A routing algorithm chooses a path for packets to reach their destination.

Example: Like a map app choosing a route.

MCQ points:

• Distance Vector shares routing information with neighboring routers.

• Distance Vector commonly uses Bellman-Ford; RIP uses it.

• Link State builds a topology map and computes shortest paths.

• Link State uses Dijkstra; OSPF uses it.

• Distance Vector can suffer from slow convergence and routing loops.

Distance Vector	Link State
Shares distance/next-hop information with neighbors.	Builds a network topology map from link-state information.
Bellman-Ford algorithm.	Dijkstra algorithm.
RIP.	OSPF.

6. Unicast Routing Protocols ■■■■■

Definition: Routing protocols let routers exchange information and select routes.

MCQ points:

• RIP = Routing Information Protocol; Distance Vector; hop-count metric; 15 usable hops, 16 means unreachable; UDP port 520.

• OSPF = Open Shortest Path First; Link State; Dijkstra; IP protocol number 89 (not TCP/UDP port).

• BGP = Border Gateway Protocol; exchanges routes between autonomous systems; uses TCP port 179.

7. Multicast Routing – Introduction ■■■■

Definition: Multicast sends data from one sender to a selected group of receivers.

Example: A live online lecture is sent to students who joined the session.

MCQ points:

• Multicast = One-to-many selected group.

• Can save bandwidth compared with separate unicast copies.

8. Multicasting Basics ■■■■

Definition: Multicast receivers join a group and receive data sent to that group.

MCQ points:

• IPv4 multicast uses Class D addresses.

• IPv4 multicast range: 224.0.0.0–239.255.255.255.

• IGMP manages IPv4 multicast group membership.

• IPv6 multicast addresses begin with FF.

9. IPv6 Addressing ■■■■■

Definition: IPv6 is the newer IP version with a much larger address space.

Example: Example: 2001:db8::1

MCQ points:

• IPv6 address = 128 bits = 16 bytes.

• Written in hexadecimal, separated by colons.

• Leading zeros in a group can be omitted.

• One continuous sequence of zero groups can be shortened to :: once.

• IPv6 supports multicast and anycast, but not broadcast.

IPv4	IPv6
32-bit; 4 bytes.	128-bit; 16 bytes.
Decimal dotted notation, e.g. 192.168.1.1.	Hexadecimal colon notation, e.g. 2001:db8::1.
Supports broadcast.	No broadcast; supports multicast and anycast.

10. IPv6 Protocol ■■■■■

Definition: IPv6 is an Internet Protocol version that uses 128-bit addresses.

MCQ points:

• IPv6 base header = 40 bytes.

• IPv4 minimum base header = 20 bytes.

• IPv6 has no header checksum.

• IPv6 uses Neighbor Discovery instead of ARP.

• IPv6 uses ICMPv6.

• Routers do not fragment IPv6 packets; source devices handle fragmentation.

UNIT V — TRANSPORT LAYER

1. Transport Layer Introduction ■■■■■

Definition: The Transport Layer provides process-to-process communication between applications on different devices.

Example: Your browser and email application use the network to communicate with different servers.

MCQ points:

• OSI Layer 4.

• Data units: TCP segment and UDP datagram.

• Port numbers identify applications/processes.

2. Transport Layer Services ■■■■■

Definition: Transport services help applications communicate.

MCQ points:

• Process-to-process delivery.

• Port addressing.

• Segmentation and reassembly.

• Flow control prevents overwhelming the receiver.

• Error control helps manage errors/loss.

• Congestion control helps avoid network overload.

• TCP provides reliability, flow control, and congestion control.

IP Address	Port Number
Identifies a device/interface on a network.	Identifies an application/process on that device.
Example: 192.168.1.10	Example: 443

3. Transport Layer Protocols ■■■■■

Definition: The main protocols are UDP, TCP, and SCTP.

MCQ points:

• UDP = User Datagram Protocol; connectionless and low overhead.

• TCP = Transmission Control Protocol; reliable and connection-oriented.

• SCTP = Stream Control Transmission Protocol; reliable and message-oriented.

4. UDP ■■■■■

Definition: UDP sends data without first establishing a connection.

Example: Like a postcard: it is sent without a delivery guarantee.

MCQ points:

• Connectionless.

• No guaranteed delivery, order, or retransmission by UDP itself.

• Low overhead and often fast.

• UDP header = 8 bytes.

• Used in gaming, streaming, voice/video, and DNS.

5. TCP ■■■■■

Definition: TCP provides reliable, ordered communication between applications.

Example: Like a tracked parcel: missing data can be retransmitted.

MCQ points:

• Connection-oriented.

• Reliable and ordered delivery.

• Uses sequence numbers and acknowledgments.

• Provides flow and congestion control.

• Minimum TCP header = 20 bytes.

TCP Three-Way Handshake ■■■■■

SYN → SYN-ACK → ACK
1. SYN: client requests connection. 2. SYN-ACK: server accepts/responds. 3. ACK: client confirms.

TCP	UDP
Connection-oriented.	Connectionless.
Reliable and ordered.	No guaranteed delivery/order.
Minimum header: 20 bytes.	Header: 8 bytes.
Web, email, file transfer.	Gaming, streaming, many DNS queries.

6. SCTP ■■■■

Definition: SCTP is a reliable, message-oriented Transport Layer protocol.

MCQ points:

• SCTP = Stream Control Transmission Protocol.

• Supports multistreaming: multiple independent streams in one association.

• Supports multihoming: one endpoint can use multiple IP addresses for redundancy.

• Uses association terminology.

UNIT VI — APPLICATION LAYER

1. Application Layer Introduction ■■■■■

Definition: The Application Layer provides network services directly to user applications.

Example: Examples: websites, email, file transfer, remote login, and DNS.

MCQ points:

• OSI Layer 7; highest OSI layer.

• Common protocols: HTTP, FTP, SMTP, DNS, SSH.

2. Client-Server Programming ■■■■■

Definition: A client requests a service; a server provides it.

Example: Like a restaurant: customer (client) orders; restaurant (server) responds.

MCQ points:

• Client initiates a request.

• Server processes it and sends a response.

• Multiple clients can communicate with one server.

3. WWW and HTTP ■■■■■

Definition: WWW is a collection of webpages/resources accessed over the Internet. HTTP is the protocol used for browser-server communication.

Example: Browser requests a webpage; web server returns a response.

MCQ points:

• WWW = World Wide Web; Internet and WWW are not the same.

• HTTP = Hypertext Transfer Protocol.

• HTTP is stateless.

• HTTP default port = 80; HTTPS default port = 443.

• HTTPS uses TLS to encrypt communication.

4. FTP ■■■■■

Definition: FTP transfers files between a client and a server.

Example: Uploading files to a website server.

MCQ points:

• FTP = File Transfer Protocol.

• Uses TCP.

• Control connection commonly uses port 21.

• Traditional active-mode data connection commonly uses port 20.

5. Electronic Mail ■■■■■

Definition: Email lets users send and receive messages over a network.

MCQ points:

• SMTP = Simple Mail Transfer Protocol; sends email.

• POP3 = Post Office Protocol version 3; commonly downloads email.

• IMAP = Internet Message Access Protocol; accesses and synchronizes mail on the server.

• SMTP commonly uses ports 25, 587, or 465 depending on configuration.

• POP3: 110 (secure 995); IMAP: 143 (secure 993).

Protocol	Main purpose
SMTP	Send email.
POP3	Retrieve/download email.
IMAP	Access and synchronize email on server.

6. SSH ■■■■■

Definition: SSH securely accesses and controls another computer over a network.

Example: A developer remotely logs into a Linux server.

MCQ points:

• SSH = Secure Shell.

• Provides encrypted remote login and command execution.

• Default port = 22.

• Often replaces insecure Telnet.

7. DNS ■■■■■

Definition: DNS translates domain names into IP addresses.

Example: You type a website name; DNS helps find its IP address.

MCQ points:

• DNS = Domain Name System.

• Common port = 53.

• Ordinary queries typically use UDP; TCP is also used in cases such as zone transfers and certain larger responses.

• A record = IPv4; AAAA = IPv6; MX = mail server; CNAME = alias; NS = name server.

DNS Record	Meaning
A	Maps a name to an IPv4 address.
AAAA	Maps a name to an IPv6 address.
MX	Mail server information.
CNAME	Alias for another domain name.
NS	Authoritative name server.

8. Compression ■■■■■

Definition: Compression reduces the size of data to save storage or transmission time.

MCQ points:

• Lossless: no information is lost; original can be recovered exactly. Examples: ZIP, PNG.

• Lossy: some information is permanently lost; often achieves greater compression. Examples: JPEG, MP3.

Lossless	Lossy
No information lost.	Some information lost.
Original data recovered exactly.	Original data not recovered exactly.
ZIP, PNG.	JPEG, MP3.

LAST-MINUTE MCQ REVISION

Topic	Remember
Network Layer	OSI Layer 3; packet; routing
Transport Layer	OSI Layer 4; process-to-process
Application Layer	OSI Layer 7; user-facing services
IPv4 / IPv6	32 bits / 128 bits
ICMP	Ping; error reporting and diagnostics
Distance Vector	Bellman-Ford; RIP
Link State	Dijkstra; OSPF
RIP	15 usable hops; 16 unreachable; UDP 520
OSPF	Link State; IP protocol number 89
BGP	Between autonomous systems; TCP 179
IPv4 multicast	Class D; 224.0.0.0–239.255.255.255
IPv6 multicast	Address begins with FF
IPv6 header	40 bytes; no header checksum
UDP	Connectionless; 8-byte header
TCP	Reliable; minimum 20-byte header
TCP handshake	SYN → SYN-ACK → ACK
SCTP	Multistreaming + multihoming
HTTP / HTTPS	80 / 443
FTP	TCP; control port 21
SMTP	Send email
POP3 / IMAP	Download / synchronize-access
SSH	Secure Shell; port 22
DNS	Domain Name System; port 53
DNS A / AAAA	IPv4 / IPv6
Lossless / Lossy	No data loss / some data loss

Exam strategy

• Memorize protocol full forms, port numbers, OSI layer numbers, header sizes, and routing algorithms.

• Focus on comparisons: IPv4 vs IPv6, TCP vs UDP, Distance Vector vs Link State, SMTP vs POP3 vs IMAP, Lossless vs Lossy.

• Watch for MCQ keywords: reliable, connectionless, shortest path, multicast, port, compression, and authentication.